

Flight Delay Contagion: Executive Memo

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To: VP of Operations, American Airlines · **Re:** Stopping delay chains at DFW

Bottom line: Don't add blanket schedule buffers at DFW; they would lose about \$20M a year. Instead, pilot buffers on the ~6,100 turns whose inbound flight is chronically late. That cuts the loss by 97% and likely pays off once knock-on delays are counted.

What we found

1. **Delays spread plane to plane, and it's costing more each year.** US airlines lost an estimated \$49.6B to delays in 2021–2025, rising from \$7.1B to \$11.7B a year. Linking 33.65M flights by tail number traced 4.1M delay chains; 41.5% of chain delay minutes were inherited from an earlier flight, close to the 38.8% late-aircraft share in BTS's own cause codes.
2. **DFW is where the costliest chains start, and American runs tighter turns than rivals.** DFW leads all airports in chain cost (\$2.88B). American has the highest chain cost of any airline (\$8.28B) and passes on 64.0% of late arrivals at DFW, vs 56.3% for Delta at ATL ($p < 0.001$). Its median turnaround at DFW is 68 minutes, vs 75 for Delta at ATL.
3. **Turnaround time absorbs delay, but blanket buffers don't pay.** Each extra buffer minute absorbs 0.57 minutes of delay (regression, 95% CI 0.55–0.60). A 75-minute rule on every DFW turn would add 1.31M ground minutes to save 121.6K delay minutes: a net loss of \$20.2M a year. Only about 18% of turns receive a late plane, so most buffer time is wasted.

Recommendation

1. **Skip blanket buffers.** A 75-minute rule on every DFW turn would lose about \$20M a year.
2. **Pilot buffers on ~6,100 high-risk turns.** Target turns whose inbound flight was late 40%+ of the prior year. Last year's lateness predicts this year's: 2025 late rates rise from 13.9% in the lowest 2024 risk tier to 36.2% in the highest. The pilot adds about 105K buffer minutes a year and cuts the loss to \$0.7M.
3. **Fix chronically late inbound flights at the source.** Buffering at DFW treats the symptom; the late inbound routes are the cause.

Impact and how to measure it

The targeted pilot breaks even when an added ground minute costs up to \$17.99, against an assumed \$24.60. It pays off if knock-on savings reach 1.37× the direct savings; the chain analysis measured 1.31× for evening delays and 2.08× for afternoon delays. Run it for one season, compare buffered turns with unbuffered ones of similar risk, and scale only if the measured multiplier clears 1.37×.

Strategy (American at DFW, 2025)	Turns buffered	Buffer min / yr	Delay min saved / yr	Break-even \$ / ground min	Net / yr
Blanket 75-min rule	74,000	1.31M	121.6K	\$9.17	– \$20.2M
Inbound late 40%+ (pilot)	6,109	105K	19.2K	\$17.99	–\$0.7M

Assumptions: delay cost of \$98.41 per aircraft minute (Airlines for America, 2025); ground-minute cost assumed at 25% of that (\$24.60); risk tiers learned on 2024 data and tested on 2025. Savings count direct delay minutes only.

Sources: BTS Reporting Carrier On-Time Performance, Jan 2021–Dec 2025 (33.65M flights); [Airlines for America, U.S. Passenger Carrier Delay Costs](#); code and models at github.com/lalithreddy335/flight-delay-contagion.